

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$\text{Mg}^{2+} \quad \frac{91}{24.3} = 3.7449$ $\text{Ca}^{2+} \quad \frac{50}{40.1} = 1.2469 \quad (1)$ ratio is 3Mg: 1Ca (1) formula must contain $4 \times \text{CO}_3$ so it is $\text{Mg}_3\text{Ca}(\text{CO}_3)_4 \quad (1)$ accept alternative correct representations e.g. $\text{CaCO}_3.3\text{MgCO}_3$ $(\text{MgCO}_3)_3.\text{CaCO}_3$		1	2	3	2	
		(ii)		concentration = $1.25 \times 10^{-6} \text{ mol dm}^{-3}$ (from calcium or magnesium concentrations) (1) moles of solid used = $\frac{220 \times 10^{-6}}{353} = 0.6232 \times 10^{-6} \text{ mol} \quad (1)$ volume = $0.50 \text{ dm}^3 \quad (1)$ allow any value in the range $0.498\text{-}0.501 \text{ dm}^3$		1 1	1	3	2	
				Question 12 total	4	3	8	15	4	9